

Multiplying and Dividing Algebraic Fractions (Homework)

Multiply.

$$\textcircled{1} \frac{x^2-16}{x+3} \cdot \frac{x^2-6x-27}{6x^3+24x^2}$$

Answers

$$\frac{(x-4)(x-9)}{6x^2}$$

$$\textcircled{2} \frac{y^2+5y+6}{y^2+y} \cdot \frac{y^3}{y^2-9}$$

$$\frac{y^2(y+2)}{(y+1)(y-3)}$$

$$\textcircled{3} \frac{b^2+2b+1}{b^2-4b+4} \cdot \frac{b^2+2b-8}{b^2-2b-3}$$

$$\frac{(b+1)(b+4)}{(b-2)(b-3)}$$

$$\textcircled{4} \frac{9x^2+45x}{9x^2-16} \cdot \frac{3x^2-2x-8}{3x^2-6x}$$

$$\frac{3(x+5)}{3x-4}$$

$$\textcircled{5} \frac{y^3+6y^2}{8y^2+16y} \cdot \frac{5y^6+20y^5+20y^4}{15y^4+90y^3}$$

$$\frac{y^2(y+2)}{24}$$

$$\textcircled{6} \frac{7b^4-175b^2}{3b+3} \cdot \frac{9b^2}{b^4+4b^3-45b^2}$$

$$\frac{21b^2(b+5)}{(b+1)(b+9)}$$

$$\textcircled{7} \frac{10-5x}{49x^5} \cdot \frac{7x^4+14x^3}{x^2-4}$$

There is more than one place to put the negative sign.

$$\frac{-5}{7x^2}$$

$$\textcircled{8} \frac{x^4-2x^2-24}{x^4-16} \cdot \frac{x^2+x-2}{5x^3}$$

$$\frac{(x^2-6)(x-1)}{5x^3(x-2)}$$

You may use a calculator

Use the negative one rule.

Factor twice.

Use the negative one rule.

$$\textcircled{9} \frac{-3x-27}{1-x^2} \cdot \frac{8x^2+8x}{2x^3+26x^2+72x}$$

$$\frac{12}{(x-1)(x+4)}$$

$$\textcircled{10} \frac{6x^2+29x-5}{3x^2+15x} \cdot \frac{30x^3}{36x^2-1}$$

$$\frac{10x^2}{6x+1}$$

$$\textcircled{11} \frac{x^2+9xy+20y^2}{x^2-4} \cdot \frac{x-2}{x^2-16y^2} \cdot \frac{10x^2+20x}{20x^4}$$

$$\frac{(x+5y)}{2x^3(x-4y)}$$

Divide.

$$\textcircled{12} \frac{b^2-12b+27}{b^2+13b+36} \div \frac{b^2-8b-9}{b^2+3b-4}$$

$$\frac{(b-3)(b-1)}{(b+9)(b+1)}$$

$$\textcircled{13} \frac{30x-30}{x-5} \div \frac{3x^2-3x}{x^2-25}$$

$$\frac{10(x+5)}{x}$$

$$\textcircled{14} \frac{8y^4}{y^2+5y+4} \div \frac{2y^2}{y^2+8y+16}$$

$$\frac{4y^2(y+4)}{y+1}$$

$$\textcircled{15} \frac{b^2-7b+10}{8b^3-800b} \div \frac{3b^4-3b^3-6b^2}{b^2-15b+50}$$

$$\frac{(b-5)(b-5)}{24b^3(b+1)(b+10)} \text{ or } \frac{(b-5)^2}{24b^3(b+1)(b+10)}$$

$$(16) \frac{-x^2 - 7x + 8}{x^2 + 5x} \div \frac{2x + 16}{x^3}$$

$$-\frac{x^2(x-1)}{2(x+5)}$$

$$(17) \frac{3y^2 + 18y}{5y^2 + 15y} \div \frac{12y^5 - 24y^4}{8y^3 + 24y^2}$$

$$\frac{2(y+6)}{5y^2(y-2)}$$

$$(18) \frac{49b^2 - 36}{3 - b} \div \frac{7b^2 + 34b + 24}{b - 3}$$

$$-\frac{7b-6}{b+4}$$

or

$$\frac{6-7b}{b+4}$$

$$(19) \frac{1}{5x^2y + 10xy^2} \div \frac{y^5}{20x^4 + 80x^3y + 80x^2y^2}$$

$$\frac{4x(x+2y)}{y^6}$$

$$(20) \frac{8x - 32}{6x + 60} \div \frac{-16 + x^2}{x^2 + 14x + 40}$$

$$\frac{4}{3}$$

$$(21) \frac{y^2 + 6y - 27}{y^2 - 81} \div \frac{y - 3}{y - 9}$$

$$1$$

$$(22) \frac{5b + 15}{-2b - 6} \div \frac{25b - 75}{2}$$

$$-\frac{1}{5(b-3)}$$

or

$$\frac{-1}{5(b-3)}$$

*
Rearrange
the terms.